

58 y/o smoker · three months of persistent cough · biopsy taken · pathology report uses three words the patient has never heard: **metaplasia, dysplasia, necrosis**

BIO 004 · WEEK 1 · CASE DEEP DIVE

Histology: The Four Tissue Types

A pathology report you can't read yet, and the tissue biology that lets you read it.

PATIENT CASE

A **58-year-old man** visits his physician after **several months of a persistent cough**. He has smoked for many years but assumed the cough was nothing serious. A bronchoscopy is performed; a small piece of tissue from the lining of one of his bronchi is removed and sent to pathology.

Several days later the pathology report arrives.

PATHOLOGY REPORT (EXCERPT)

Specimen: right mainstem bronchus, mucosal biopsy. Microscopic examination reveals areas of *squamous metaplasia*. Regions of *dysplasia* are identified. Focal areas of *necrosis* are noted. The normal pseudostratified ciliated columnar epithelium has been largely replaced.

For most patients, those words are meaningless. For many beginning anatomy students, they're not much clearer. By the end of this page they will read like a sentence in plain English, because the language of pathology is built directly on the language of normal tissue.

GUIDING QUESTION

What is the normal tissue lining a bronchus, what happens when smoke insults that lining for thirty years, and what is the word for each step of the change?

This is the case-based companion. Terms, video, and spaced recall for the four tissue types, where tissues come from, epithelial, connective, cartilage and bone overview, muscle and nervous overview, and body membranes live on your Week 1 prework page.

1 WHERE TISSUES COME FROM

Three germ layers, every tissue in the body

Every tissue in your body started as one of three sheets in the early embryo: **ectoderm** (outer), **mesoderm** (middle), and **endoderm** (inner). These three sheets fold, migrate, and differentiate into every cell type you have.

Ectoderm

Epidermis of skin and its appendages; nervous tissue (brain, spinal cord, peripheral nerves); enamel of teeth; anterior pituitary; adrenal medulla; lens of the eye.

Mesoderm

Most connective tissues (bone, cartilage, blood, dermis); all muscle; cardiovascular and lymphatic systems; urogenital system; serous linings.

Endoderm

Epithelial lining of the GI tract and respiratory tract (including the bronchial lining our patient lost); liver; pancreas; thyroid; bladder lining.

Our patient's bronchial epithelium is *endodermal* in origin. Knowing where a tissue came from also tells you what it can become if it's pushed to adapt (metaplasia) and what it cannot.

Every part of you, built from four building blocks

However many trillions of cells you have, they are organized into exactly four primary tissue types. Memorize this list. The rest of histology is variations on the theme.

● **Epithelial tissue**

Covers surfaces and lines cavities; forms glands.

Cells packed tightly, sit on a basement membrane, avascular. Job: protection, absorption, secretion, filtration.

● **Connective tissue**

Binds, supports, protects, transports. Cells

embedded in extracellular matrix. Most diverse category by far: bone, cartilage, blood, adipose, dense and loose CT.

● **Muscle tissue**

Generates force through contraction of actin and myosin. Three subtypes: skeletal (voluntary, striated), cardiac (involuntary, striated), smooth (involuntary, non-striated).

● **Nervous tissue**

Transmits and processes electrical signals. Two cell types: neurons (signaling) and neuroglia (support).

Classified by shape, layers, and what sits on top

Epithelial tissue is classified by two features: **number of layers** (simple = one layer; stratified = multiple) and **cell shape** (squamous = flat, cuboidal = box, columnar = tall). Mix and match: *simple squamous*, *simple cuboidal*, *simple columnar*, *stratified squamous*, and so on. Two specialty types complete the list: *pseudostratified* (looks layered but isn't, all cells touch the basement membrane) and *transitional* (changes shape with stretch, lines the urinary bladder).

The lining of the bronchus that we want to know about in our patient is **pseudostratified ciliated columnar epithelium**. The cells are tall and reach the basement membrane; some look short because their nuclei sit lower; the apical surface carries **cilia** (motile hair-like projections) that beat in coordinated waves, sweeping mucus and trapped particles upward toward the throat. Interspersed between the ciliated cells are **goblet cells** that secrete the mucus the cilia sweep. Together, this is the *mucociliary escalator*, and it is what keeps your lungs clean every minute of every day.

What smoke does to the escalator. Cigarette smoke paralyzes and eventually destroys cilia. Goblet cells overproduce mucus in response. Mucus accumulates because it can't be swept up. That accumulation is what the cough is trying to clear.

TYPE	WHERE	JOB
Simple squamous	alveoli, capillary endothelium, serous linings	diffusion across a thin barrier
Simple cuboidal	kidney tubules, gland ducts	secretion and absorption
Simple columnar	most of the GI tract	absorption (often with microvilli)
Pseudostratified ciliated columnar	trachea, bronchi, upper airway	mucociliary escalator
Stratified squamous	epidermis, oral cavity, esophagus, vagina	protection against abrasion
Transitional	urinary bladder, ureter	

TYPE	WHERE	JOB
		stretch with filling and emptying

4 CONNECTIVE TISSUE

Cells scattered in a matrix that does the structural work

Connective tissue is the most diverse category. The defining feature is that the **extracellular matrix** (the goo and fibers BETWEEN cells) is what carries the load, not the cells themselves. Cells are usually sparse. The matrix is the structure.

Two components define every connective tissue: **ground substance** (the gel-like fluid background) and **fibers** (collagen for strength, elastic for stretch and recoil, reticular for delicate scaffolding). The proportions and the fiber types tell you which connective tissue you're looking at.

 Loose connective (areolar) Mixed cells, loose fibers, lots of ground substance. Wraps organs, fills spaces, supports epithelia.	 Adipose Specialized cells (adipocytes) storing fat. Insulation, energy storage, cushioning.
 Reticular Reticular fibers forming delicate meshworks. Stroma of lymph nodes, spleen, bone marrow.	 Dense regular Parallel bundles of collagen. Tendons (muscle to bone) and ligaments (bone to bone).
 Dense irregular Collagen in all directions. Dermis of skin, organ capsules.	 Cartilage Chondrocytes in a firm matrix. Hyaline, elastic, fibrocartilage. Covered separately on the Bone Microanatomy page.
 Bone Osteocytes in a mineralized matrix. The Bone Microanatomy page walks the cells and the osteon.	 Blood Fluid matrix (plasma) with formed elements (red cells, white cells, platelets). Yes, blood is a connective tissue.

Two tissues that move and signal

MUSCLE

TYPE	STRIATED?	CONTROL	WHERE
Skeletal	yes	voluntary	attached to bones, moves the skeleton
Cardiac	yes	involuntary	walls of the heart only
Smooth	no	involuntary	walls of hollow organs (gut, vessels, bladder, uterus, airways)

All three contract through the interaction of **actin** and **myosin** filaments. The difference is how those filaments are organized (striated = highly ordered into sarcomeres; smooth = less ordered) and how contraction is controlled (voluntary motor nerves vs autonomic vs intrinsic pacemaker activity).

NERVOUS

Two cell types: **neurons**, which receive and transmit electrical signals; and **neuroglia** (glial cells), which support, insulate, and protect neurons. The brain is roughly half neurons and half glia. The peripheral nervous system uses Schwann cells; the central nervous system uses astrocytes, oligodendrocytes, microglia, and ependymal cells.

How tissues combine, and what happens when they're insulted

BODY MEMBRANES

A *membrane* in this context is a thin sheet built of two tissue types (epithelium plus connective tissue) that lines or covers a body surface. Four kinds:

Cutaneous (skin)

The integument. Stratified squamous keratinized epithelium plus dense irregular connective tissue.

Mucous

Lines body cavities open to the exterior (GI, respiratory, urinary, reproductive tracts). Epithelium varies by location plus loose CT (lamina propria).

Serous

Lines body cavities closed to the exterior (pleura, pericardium, peritoneum). Simple squamous epithelium (mesothelium) plus thin CT.

Synovial

Lines joint cavities. Specialized connective tissue (not epithelium at all). Secretes synovial fluid.

TISSUE ADAPTATION, DYSPLASIA, AND THE PATH TO CANCER

When a tissue is insulted for a long time, it adapts. There is a vocabulary for each step, and the pathology report reads like a list of those steps:

Hypertrophy

Cells grow larger. Normal in pregnancy, exercise; abnormal in untreated hypertension.

Hyperplasia

Cells grow more numerous. Endometrium during cycle, BPH in older men.

Metaplasia

One mature tissue type is replaced by another mature tissue type. Reversible if the insult stops.
Squamous metaplasia of the bronchus:
pseudostratified ciliated columnar replaced by stratified squamous, because the squamous epithelium tolerates smoke better. But it has lost its cilia and its mucus production.

Dysplasia

Disordered growth. Cells of varying size, shape, organization. Not yet cancer, but the precursor lesion. Often reversible if the insult stops.

● **Neoplasia (cancer)**

Uncontrolled, autonomous growth. Cells no longer respond to the normal "stop" signals. Can invade and metastasize.

● **Atrophy**

Cells shrink or are lost from disuse, denervation, or loss of blood supply. Muscle wasting after a cast comes off.

● **Necrosis**

Pathologic cell death from injury (ischemia, toxin, infection). Cells swell and burst, releasing contents that trigger inflammation.

● **Apoptosis**

Programmed, orderly cell death. Cells shrink and break into membrane-bound fragments that are quietly cleaned up without inflammation. Normal turnover, development.



DRAW ZONE 1 -- *Sketch the four tissue types side by side. Label one example of each.*

Sketch epithelial (use simple columnar), connective (use loose areolar), muscle (use skeletal), nervous (use a neuron).

 **DRAW ZONE 2** -- Sketch the bronchial lining in four panels: normal pseudostratified ciliated columnar → squamous metaplasia → dysplasia → neoplasia.

Mark cilia and goblet cells in panel 1. Lose the cilia in panels 2-4. Show disorganization in 3 and 4.

Read it now, in plain English

The three words that meant nothing on the first read now name three distinct stages of tissue adaptation under thirty years of insult.

"*Squamous metaplasia*"

The normal **pseudostratified ciliated columnar epithelium** of the bronchus has been replaced by **stratified squamous**. Squamous tolerates the chemical and thermal insults of smoke better. Cost: cilia gone, mucociliary escalator gone. The cough is now the lungs' only way to clear secretions.

"*Dysplasia*"

Cells in the new squamous epithelium are growing in disordered fashion: varying size, varying shape, abnormal arrangement. Not yet cancer, but on the path. The cell cycle is starting to escape normal control.

"*Focal necrosis*"

Patches of cells have died from injury rather than orderly turnover. Dead cell contents leaked into the tissue and triggered local inflammation.

"*What's next on this trajectory?*"

If the insult continues, dysplasia may progress to *carcinoma in situ* and then to invasive *squamous cell carcinoma*. If smoking stops, mild and moderate dysplasia can regress; severe dysplasia usually does not.

"*Why the cough won't stop*"

Cilia are gone (metaplasia). Goblet cells are overproducing mucus. Excess mucus stays in the lower airway because there's no escalator to clear it. Cough is the only mechanism left.

"*Is the damage reversible?*"

Metaplasia is largely reversible if smoking stops; airway cilia and goblet cells can recover over months. Dysplasia is partially reversible. Invasive cancer is not.

Normal, adapted, disordered, dead. Four words for what happened in this patient's bronchus over thirty years. When you go back to the prework, the slides of epithelium and connective tissue will land differently. Each one is now connected to a patient whose tissue is telling you a story.

